

Examiner reports

Q1.

Part (a) split students into two groups. Almost equal numbers scored full marks or no marks. Finding the gradient of a curve given in parametric form should be a well-practiced and routine technique. However, differentiating 2^t is new to this specification and this caught out many students.

Part (b) was completed more successfully. Students who took an approach similar to the typical solution given in the mark scheme, tended to complete the question and often scored full marks. Students who introduced logs, to make t the subject, scored the first mark for choosing a correct strategy. These students were often unable to manipulate their logs and failed to complete the argument correctly, which is what the third mark was for.

Q3.

This question was not done well, with fewer than 10% of the entry gaining full marks, although over half did gain 4 or more of the 8 marks available.

Part (a)(i). Most students were successful here; using implicit differentiation as expected and clearly showing the derivatives of the three terms. Those few who solved y in terms of x before differentiating were rarely successful, particularly those who thought that $y = x - \sqrt{8}$. Some students left their derivative in terms of x and y rather than the requested p and q .

Part (a)(ii). Many students gained the first 2 marks for a correct expression for the tangents, especially those who used $y - y_1 = m(x - x_1)$. The relatively few who tried to use $y = mx + c$ often made an error in finding c in terms of p and q . The common mistake was a sign error in the gradient at $(p, -q)$, reducing the attempt at solving the two tangent equations simultaneously to nonsense. The simplest way to establish the requested result is to add the two tangent equations, but few students did this. Most equated the two expressions for y and attempted to find x , often making an algebraic error. This is a valid method if the expression for x is substituted back in a tangent equation and $y = 0$ is shown, but few attempted to do this. Other students assumed $y = 0$ and attempted to find an expression for x but most were unsure how to proceed from there. Other students wrote their tangent equations in the form $ax + by + c = 0$ and equated them both to zero. Again it is possible to solve the problem this way, although most students could not complete successfully.

Part (b). A variety of attempts at this question were seen, including those who thought they had been asked to find the derivative using the parametric equations, which were irrelevant. Some students seemed not to understand the question at all. Those who squared the expressions for x and y and subtracted were usually successful, although many made an error in the squaring by missing a term or making a sign error. Very few factorised $x^2 - y^2$ and then used the parametric equations to eliminate t , but some students did establish the given result by other algebraically valid eliminations of t .

Q4.

This question was generally done well with over 50% of the entry scoring 9 or more of the 12 marks available. Most students lost marks in part (c) by attempting to use an inefficient method.

- (a) Virtually all students used parametric differentiation and most had the two required derivatives correct. Occasionally there was a coefficient error or a t was brought down, but most students knew the chain rule and used it appropriately. Most students had the correct unsimplified result, which was accepted for the marks, but some made an error in simplifying, usually dropping the negative sign and so lost two marks in part(b). Some students 'simplified' $1 / 16e^{-2t}$ to $16e^{2t}$.
- (b) (i) Most students had this correct, either by use of their calculator or by hand, although there was evidence that some students were not using the exponential button on their calculator correctly.
- (ii) Although again most students had these coordinates correct, a number dropped the 4 from one of the given equations, usually from the x equation, so lost a mark here and another in part (iii) as they could not then get the correct answer.
- (iii) Most students used their results from parts (i) and (ii) to set up a correct equation of the normal, either using $y - y_1 = m(x - x_1)$ or $y = mx + c$. Some went on to complete the question successfully but others confused the axes and substituted $x = 0$ and found y . Some students misunderstood the question and substituted $y = 0$ into the parametric equation for y and found t , seemingly ignoring the reference to the normal given in the question.
- (c) There were many different approaches to finding the Cartesian equation. The most straight forward way was to substitute into the given expression and show that it simplified to the requested constant. The common error in this approach was a sign error, often resulting in $k = 0$. The result $e^{2t} e^{-2t} = 1$ was generally well known although occasionally the product became $e^{\pm 4t}$. The other fairly common approach was to solve one of the equations for $e^{\pm 2t}$ and substitute in the other equation, and then simplify. This often resulted in algebraic errors. Some students carried out an unnecessary further stage and actually solved for t in terms of x and / or y using logarithms. This often resulted in some rather cumbersome expressions when substituting back and students made many sign and algebraic errors in trying to convert their logarithm expressions into the requested form.

A typical error was $e^{-\ln\left(\frac{x+4}{8}\right)} = -\left(\frac{x+4}{8}\right)$. Some students did however derive the required result through insightful manipulation of the parametric equations.

Q5.

Most of the candidates had part (a) correct, with the majority choosing to substitute $t = \frac{3}{y}$ into the given equation for x rather than to substitute into the given cartesian equation. The commonest, but rarely seen, error was to not square the 3 or to make a sign error.

The majority of candidates chose to approach part (b)(i) through parametric differentiation, and most did it accurately, with many gaining full marks. The most common error was to

lose the minus sign on the -16 gradient, having initially found this correctly. Others made mistakes with the fractions.

For part (b)(ii), there are numerous ways in which this intersection can be verified, and many different ways were attempted by the candidates. Many scored 1 mark by finding the y - value, either directly from their tangent or from the curve, or even via the parametric equations.

Some substituted their tangent equation into the curve and attempted to verify that $x = \frac{3}{2}$ satisfied the equation. Some did this convincingly, but others proceeded rather more in hope than expectation. Some tried to factorise the resulting cubic equation. The most efficient method was to find the y -value from the equation of the tangent and then verify

that $(\frac{3}{2}, -8)$ satisfies the equation of the curve. This method was mostly seen from those who achieved full marks on the question.

Q6.

For part (a)(i), most students knew that they needed to differentiate the two parametric equations and use the chain rule which many did correctly, with very few having it the wrong way up. Errors were relatively rare but included sign errors, coefficient errors or dropping the 2 from the double angle. Some chose to expand $\sin 2\theta$ and use the product rule, again mostly successfully, but some made sign errors here too.

The manipulation to the required form was often confused as some students lost their way, many of these immediately replacing the $\frac{1}{\sin \theta}$ term from differentiation by $\operatorname{cosec} \theta$, and abandoning, or fudging, as they could not get to the required form. Those who immediately replaced $\cos 2\alpha$ with $1 - 2 \sin^2 \alpha$ usually went on to be successful. The common error was to omit the 2 in the expansion. Those who went via the $\cos^2 \theta - \sin^2 \theta$ form were sometimes successful, although many made a sign or coefficient error in attempting to simplify to the required form.

Part (a)(ii) was essentially testing a student's ability to use $\theta = \frac{\pi}{6}$ correctly in their version of $a \sin \alpha + b \operatorname{cosec} \alpha$ in finding the gradient of a tangent and the associated normal. Many did it correctly, although a significant number only found the gradient of the tangent whereas some unnecessarily found the equation of the normal. Some students used their chain rule expression, which was perfectly acceptable. The relationship between the gradients was generally very well known.

Many students made little or no progress with part (b), especially if they tried to start from inverse sines and cosines from the parametric equations. Those who expanded $\sin 2\theta$ and then squared the whole equation usually progressed to trying to use the parametric equations to eliminate θ , but relatively few could then manipulate correctly to the required form. Few scored high marks here, but some very insightful and clever derivations were seen, based on $\sin^2 \theta + \cos^2 \theta = 1$.

Q7.

This question was also done well, with again nearly half the entry gaining full marks.

In part (a)(i) most candidates found the derivatives $\frac{dx}{dt}$ and $\frac{dy}{dt}$ successfully and went on to

divide to find $\frac{dy}{dx}$ correctly. The most common, although rare, error was the omission of the

minus sign when differentiating e^{-2t} . Occasionally a candidate used $\frac{dx}{dt} / \frac{dy}{dt}$.

In (a)(ii) almost all candidates gained this mark for a correct tan gent equation using their numerical gradient from part (a)(i). Only a few took the negative reciprocal to find the equation of the normal.

In part (b) most candidates were successful too. Whilst some used the quickest method of

substituting $e^t = \frac{x}{3}$ or $e^{2t} = \frac{x^2}{9}$, others used logarithms. Candidates were required to conclude with the correct equation, which some failed to do; a verification was given no credit.

Q8.

Most candidates scored the marks for the two derivatives with relatively few omitting the minus signs. Some candidates managed to drop the 2 from $\cos 2\theta$. Many candidates brought the

commonly used parameter t into their expression using $\frac{dx}{dt} = -\sin \theta$ or similar, but were not penalised. Virtually all knew how to use the chain rule, but many got mixed up with the coefficients in both expanding $\sin 2\theta$ and simplifying the expression. The common error was to find $k = 3$. Some candidates got a result in terms of $\sin \theta$ and some got no result at all, but any value of k , however found or invented, was allowed in part (a)(ii).

Part (a)(ii) was generally done well with most candidates demonstrating they both knew and could use the relation between the gradients of the tan gent and the normal, although a few

found an equation of the tan gent. The evaluation using $\theta = \frac{\pi}{3}$ was generally accurately done; however candidates had to have derived $k = 6$ correctly to gain the final mark for the equation of the normal.

In part (b) it was expected here that candidates would note the whole question was about the use of double angle formulae, and many did. Most got as far as at least trying to express $\sin^2 x$ in term of $\cos 2x$ and of those, many did it correctly, even if sometimes there was a mix of variables. The integration of $\cos 2x$ was usually accurate, with the common error being to double $\sin 2x$ rather than halve it. There were few sign errors. Virtually all candidates who achieved an

integral knew how to use the limits, but the real test here was in handling $\sin 2\left(-\frac{\pi}{4}\right)$; many candidates lost credit through not showing their evaluation or getting the sign wrong.

However, many candidates, about 25% of the entry, went on to obtain the correct answer and give it in an exact form, although some did make a mistake with the signs, particularly the three minus signs involved with the lower limit. Of the few candidates who attempted the integral using parts most made little progress beyond the first line of working, not realising that in order to proceed with the integral they needed to change $\cos^2 x$ back to $\sin^2 x$, and manipulate the resulting expression.

Q9.

In part (a)(i), most candidates stated the formula for $\sin 2\theta$ correctly although not always in its simplest form.

In part (a)(ii), although many candidates knew the formula for $\cos 2\theta$ in at least one its correct forms, several gave $\cos^2 \theta + \sin^2 \theta$, and later in the question just gave the value of $\cos 2\theta$ as 1 without apparently thinking something could be wrong.

In part (a)(ii), many candidates had problems. It appeared that the 3-4-5 triangle was either not well known or not recognised as relevant; although some candidates did draw this to justify their answer others used $\sin^2 \theta = 1 - \cos^2 \theta$. Many candidates worked backwards from the answer given, to find a value for $\sin \theta$, which gained no credit nor did going via an angle from the given value of $\cos \theta$. Many candidates omitted to find $\cos 2\theta$ or, as above, gave the value as 1, or another incorrect value from an incorrect expression. Those who did find the correct value for $\cos 2\theta$ often dropped the minus sign in later working.

In part (b)(i), the use of the chain rule with these parametric equations was generally done well with most candidates completing it correctly. Some candidates apparently tried to integrate rather than differentiate and some made sign errors when differentiating. A few candidates had the chain rule expression the wrong way up.

In part (b)(ii), although some neat and concise correct answers were seen, many candidates got in a mess trying to use angles and not working in exact fractions. Some confused the value of the sine and cosines involved with the angles writing terms such as $\sin \frac{3}{5}$. Some left a mix of fractions and decimal numbers in their final answer, which is bad practice but was accepted if the values of the coordinates at P were correct.

Q10.

- (a) The vast majority of candidates completed this question successfully and as expected, by finding the two derivatives with respect to t and combining them using the chain rule. The more common errors were to omit the squared powered from $6t^2$ or the negative sign in -3 . Some candidates lost a mark through not simplifying their final result. Relatively few candidates made the error of multiplying the derivatives or having the chain rule upside down. Very few candidates took the approach of eliminating t and seeking an equation $y = f(x)$. Of those who did, hardly any were successful in differentiating their expression.
- (b) The vast majority of candidates was successful here too, substituting $t = 1$ immediately and obtaining the gradient of the normal correctly and then a correct form of the equation of the normal. Those few candidates who made no attempt to

find the gradient of the normal and just used the tangent gained no credit. The fairly rare error in finding the gradient of the normal was to omit the negative sign. Some candidates also failed to substitute for the given value for t and gained no credit.

- (c) Various correct approaches were seen with the most efficient being to obtain t in terms of x and substitute into $y = f(t)$, and many candidates did this correctly. The common errors were to omit or have an unclearly positioned negative sign, or to make an error in the power. Other attempts at elimination were generally not so successful. Many candidates wasted time through proceeding to expand their cartesian equation, or just generally and unnecessarily manipulating it algebraically. Once t is eliminated, the resulting equation is a cartesian equation.

Q11.

In part (a) almost all candidates obtained $dx/dt = -1/t^2$. However, most candidates could not handle the fraction and indices correctly when finding dy/dt . The most common mistake was to state $dy/dt = 1 - 2/t^2$. Sometimes this “became” $1 - 1/2t^2$ but, by then, the mark had been lost. The t^2 sometimes migrated on to the numerator. Almost all candidates knew they needed to divide dy/dt by dx/dt to get dy/dx . Full marks were awarded for an unsimplified form of dy/dx .

A few candidates confused differentiation and integration giving the “derivatives” in terms of $\ln t$. Some candidates proceeded by first attempting to find the cartesian equation, which was valid if the result was given in terms of t . Surprisingly many of these candidates then didn’t make much progress in part (c). In part (b) the majority of candidates found the equation of the normal, and not the tangent, and the relationship between the gradients was well known.

Very few found an incorrect y value at $t = 1$. Most errors were due to an incorrect dy/dx from part (a), but some credit was given for method and 3 marks were still available. These were earned by most. Some candidates had an expression for dy/dt in which the t ’s cancelled, yet it didn’t apparently occur to them that an error had been made. Most candidates began by writing the cartesian equation in an unsimplified form in part (c). Many were defeated by the fraction in the denominator and so could not cope with $1/(2(1/x))$. Errors were often made when attempting to multiply through by $2x$. Some candidates chose to substitute for t in the given equation, but not all could deal with the algebra involved; $k = 1$ or $k = -2$ were common wrong conclusions. Pleasingly most candidates who did attempt this question made an algebraic approach and very few just substituted $x = 1$ and $y = 3/2$ to find the value of k , which gained no credit.

Q12.

Most candidates were correct in part (a)(i), with very few arithmetic errors seen. Similarly, the differentiation required for part (a)(ii) was usually done correctly, with the common

error being $\frac{d}{dt} \left(\frac{1}{t^2} \right) = -\frac{1}{t}$. Most candidates used the chain rule correctly, although some

poor algebra was seen in inverting $\frac{dx}{dt}$, with $\frac{1}{2 - 2t^{-3}} = \frac{1}{2} - \frac{1}{2t^{-3}}$ being seen quite often; if candidates had not made it clear that this is what they believed the reciprocal to be, they

were denied the chain rule method mark. However, most candidates scored the mark for finding a tangent, although a few found an equation for the normal.

Those who chose to find c in $y = mx + c$ instead of using $(y + 3) = m(x - 5)$ were prone to error. Part (b) was much less successful for most candidates. Despite the required form of the answer being given, few candidates chose to find $(x + y)$ and $(x - y)$ in terms of t and apply the result to the given equation. Of those that did do this, there were many errors in the squaring and multiplication. Some candidates tried to find t in terms of x and y , often getting lost algebraically in their attempt and abandoning it; some did manage to complete this method successfully, but it was time consuming. Others simply did not attempt this part of the question.

Q13.

This was the most successful question on the paper with most candidates gaining full marks or close to it. Simple sign and coefficient errors were made in the differentiation, but the vast majority of candidates demonstrated they could use the chain rule correctly. Only a few tried to find y in terms of x first, doing the work actually required for part (c). Some candidates found an equation of the tangent instead of the normal, and a few left t in their final answer. A form of the answer was not specified so any correct form was accepted, but some working with the $y = mx + c$ form sometimes found a value for c , but did not write down an equation as their final answer.

Similarly in part (c) a correct, unsimplified form was accepted for full credit, although it was noted that many did go on to make algebraic errors in attempting to simplify their equation. Most candidates did know what was required for a cartesian equation and many alternative correct versions were seen or a simple sign error was made. A few candidates thought they were required to find the tangent, and some seemed to invent their own equation, such as an ellipse, to substitute into.

Q14.

Most candidates differentiated the functions in part (a) correctly, but with some making sign or coefficient errors. Most also used the chain rule correctly, with relatively few getting it upside down or as a product. Evaluation was not always successful, with the common error being to have the calculator in degrees mode when working with radian measure.

Answers in part (b) varied from no attempt to very well presented clear demonstrations of the result with $k = 4$. Curiously, some well-presented solutions had $k = 2$, with candidates omitting to square this term.

Many candidates got themselves into difficulties through trying to recollect and manipulate trigonometric identities, with little apparent thought as to where they were heading, often making algebraic errors. Their work was often difficult to follow. Other candidates showed a misunderstanding of the question in that they gave a value for k from their working, without finding the whole expression. Some candidates got confused between the variables x and θ , with some just using $\cos$ or $\sin$ with no variable attached.

Q15.

The great majority of candidates knew what was expected in part (a) but some made

errors in attempting $\frac{d}{dt}\left(\frac{2}{t}\right)$; sign and coefficient errors being common and even $\ln t$ was seen.

The chain rule was usually used correctly although a few multiplied the two derivatives or had the expression 'upside down'. Errors were sometimes made in simplifying the expression, with a sign error or 2 or 8 being seen instead of $1/2$, or the t^2 term moving to the top line.

Candidates generally showed they knew how to find the equation of a straight line in part (b) using their gradient from part (a) although a few found the normal rather than the tangent. Some candidates lost a mark through not giving their final answer in the requested form, or did not notice that their a , b , c were not integers, particularly those few who had a $\ln$ term in part (a). Part (c) was usually started well but not always finished convincingly, as the answer was given and sufficient working not shown. The question did request a verification of the given answer rather than requiring it to be shown, but few

candidates did it by the most efficient method of demonstrating $-4t \times \frac{2}{t} = -8$. There were many good answers using the elimination of t , although some candidates were too quick to write down the given answer without showing fully where it had come from.

Q16.

Part (a) The vast majority of candidates calculated the coordinates correctly.

Part (b) There was a mixed response with some candidates confidently handling the algebra and others getting into difficulties in their attempt to eliminate t . Some verified the given equation by substituting the given expressions for x and y which gained partial credit, but no credit was given for verification using (3, 2) and claiming that the equation works.

Part (c) There were essentially three approaches as most candidates realised they had to do some form of differentiation. These were using parametric and implicit differentiation to find dy/dx or solving for x and finding dx/dy ; the latter was usually successful. The use of the two expected techniques was very mixed, some candidates confidently and clearly finding the derivatives and using $t = 1/2$ or (3, 2) to obtain the answer given. Others were far less sure of what they were trying to do. There were errors in the chain rule such as

$\frac{dx}{dy} = \frac{dy}{dt} \frac{dx}{dt}$ in the derivatives with $\frac{d}{dt}\left(\frac{1}{t}\right) = \ln t$ being seen quite often. There were also errors in attempted simplifications of the algebraic fraction often resulting in a candidate just writing down gradient = 2. Those who used implicit differentiation often had a loose dx/dy at the front of their expression, which sometimes stayed in it. Errors were more common with the product xy than in the y^2 term. Some candidates who had the derivatives correct again just wrote down the given gradient, some just writing **calculator** beside their answer; they needed to show where 2 came from to gain full credit.